Iteration #2

Section 3: System Design
Main ->

- **User creates chooses or creates a new *UserProfile***
  - ***UserProfile* -> needs a "Name" and "Age"**
- **From the *UserProfile*, a new *Reading* is created for the *UserProfile***
  - ***Reading* needs an "Activity"**
- **A new *HeartRateSensor* instance is created which stores the times at which a "HeartBeat" is detected. The *HeartRateSensor* can be started or stopped.**
- **User inputs "q" to quit/stop program. Data is then processed:**

3.1 Classes

UserProfile: the person wearing the tracker

Private: string name, int age, int restingHR
Public: setProfile(name, age, restingHR), getName(), getAgee(), getResingHR(), getMaxHr() (220-age), getZone(bpm) (returns zone name), print()

HeartRateSensor: simulates the watch sensor

Private: ReadingLog readings
Public: start(), stop(), getAverageBPM(), getRecentAverageBPMFrom(float seconds), getLastAverageBPM(int numberOfRecentReadings)

ReadingLog: stores all readings

Private: int[] readings
Public: add(Readings), size(), get(index), minBpm(), maxBpm(), avgBpn(), print(), printLast(n), clear()

AlertSystem: checks for dangerous values (200-age is the max)

Private: int highlimit, int lowlimit
Public: setLimits(low,high), isAlert(Reading), printAlerts(ReadingLog)

FileManager: saves and loads data

Private: string filename
Public: save(UserProfile, ReadingLog), load(UserProfile&ReadingLog)

Menu Design:

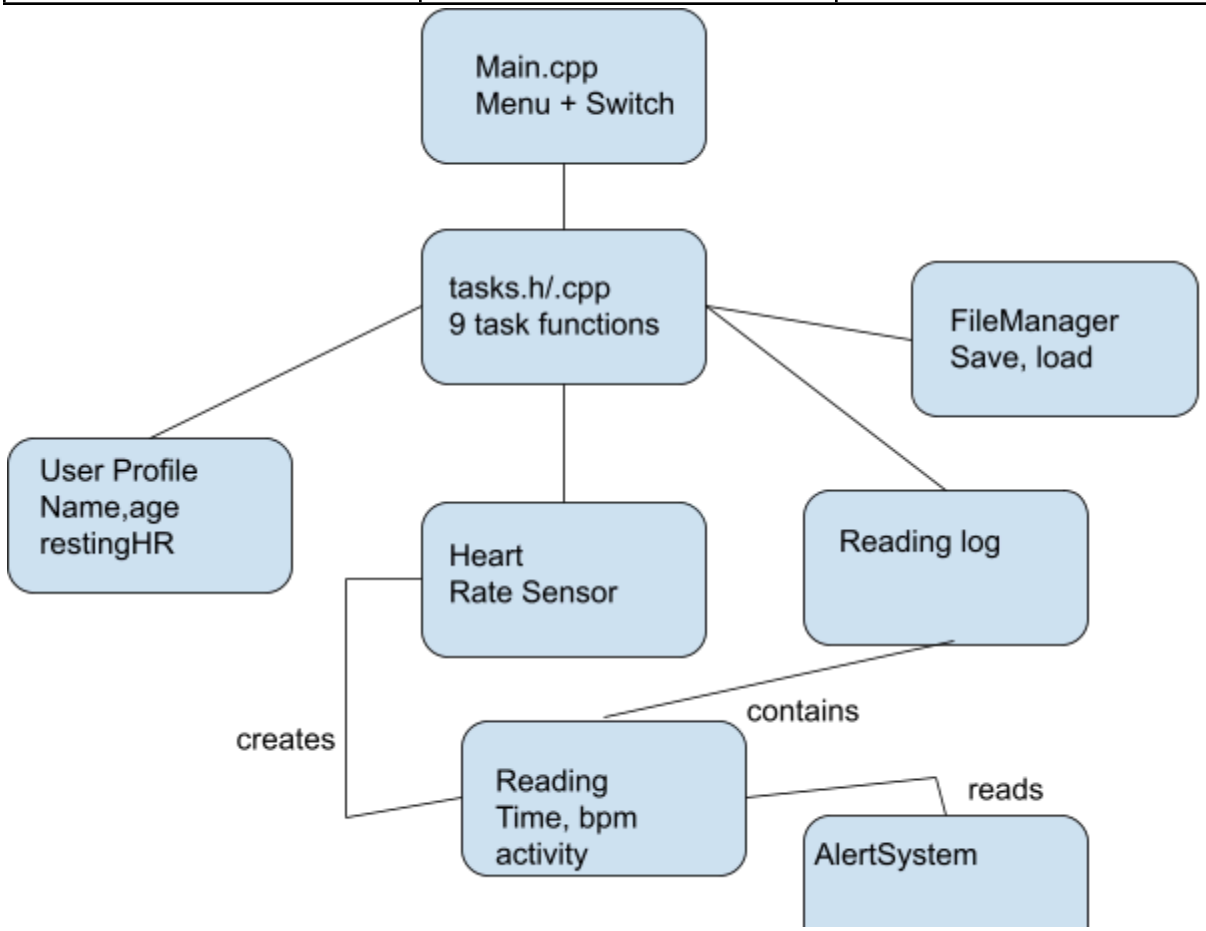
1. Create/Select Profile → taskCreateProfile()
2. Start workout and input activity → taskRunWorkout()
3. View latest readings —> tastViewLatest()
4. Saves data —> taskSave()
5. Load data —> taskLoad()
6. View full history —> taskViewHistor()
7. Statistics —> taskStatistics()
8. Heart rate zones (HRR) —> taskZoneSummary()
9. Alerts —> taskAlerts ()
   0. Exit

main() reads the choice and uses a switch with one case per option, each calling one function

File Structure

Files	Contains	Owner
main.cpp	Menu, switch, creates the shared objects	B
tasks.h	Declaration of all nine task functions	Shared
tasks_data.cpp	Task 1 to 5	A
tasks_analysis.cpp	Tasks 6 to 9	B
readings.h/.cpp	Reading	A
User .h/.cpp	UserProfile	B
Sensor .h/.cpp	HeartRateSensor	A
Filemanager .h/.cpp	FileManager	A

readingl.h/.cpp	ReadingLog	B
alert.h/.cpp	Alert System	A



Iteration #2 Contributions:

Vivian	Kai
1. Defining all tasks 2. Design the system ideas 3. Creating main such as cases and menu	1. Userprofile 2. Started HeartRateSensor 3. Modified system ideas